PROJECT PLAN: Digital Examination Attendance Verification System

Team Members:

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Project Duration: 6 Weeks

1. Introduction

Background

The University of Eswatini currently uses physical examination cards to verify student attendance during examinations. These cards are signed by students and collected by invigilators as proof that a student sat for an examination.

However, this manual system of verifying student attendance presents problems such as:

- Loss or misplacement of examination cards
- Error-prone manual counting of students
- Lack of secure and traceable attendance records

Aim

The aim of this project is to develop a digital examination attendance verification system that can verify that a student has sat for an exam, confirm that the person writing the examination is the actual registered student, track that the student submitted their examination script and provide a total sum of students that has sat for each courses exam. This will improve efficiency, security, and reliability.

Objectives

The project aims to digitize examination attendance verification, confirm that the person writing the examination is the actual registered student, track the submission of examination scripts and automate student attendance counting.

2. User/Client Involvement

Users/Clients

- Students
- Invigilators/ Lecturers
- Examination Markers

Client Contributions

The client will provide:

- Student registration records (Week 1)
- Current exam attendance procedures (Week 1)
- Access to sample examination schedules (Week 2)
- Feedback during prototype testing (Week 5)

3. RISKS

Risk	Impact	Mitigation
System development delays	High	Weekly milestones
Insufficient requirements	Medium	Early user consultation
Team coordination problems	Medium	Weekly progress meetings
Limited time (6 weeks)	High	Limit scope to prototype

4. Standards, Guidelines and Procedures

The team will follow:

- Structured Software Development Life Cycle (SDLC)
- Coding standards for consistent programming
- Version control using Git
- Weekly team meetings
- Proper documentation procedures
- University ethical and academic guidelines

5. Organization of the Project

Roles

Project Leader

The project leader will coordinate the projects activities, track the teams progress and ensures that deadlines are met.

Designer

The designer will create the interface design and produce the system models and diagrams.

Programmer

The programmer implements system functionality.

Tester

The tester tests system components and reports defects.

6. Project Phases

Development Model

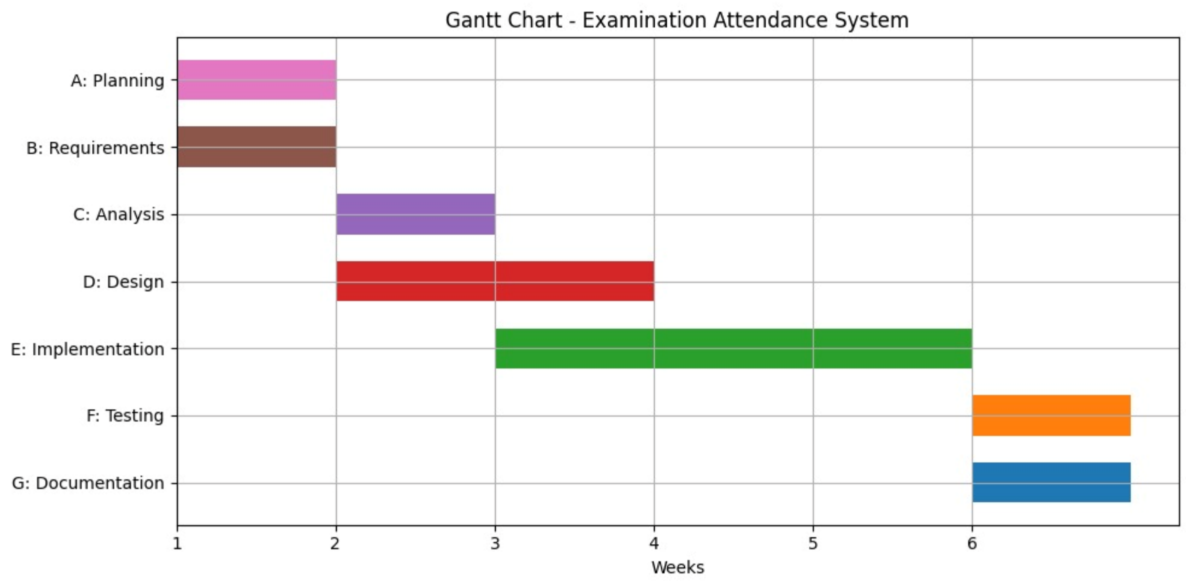
The project will adopt the Incremental development model. The system will be developed in small functional increments, with each increment being refined through multiple iterations. This approach allows flexibility, continuous improvement, and early delivery of working components.

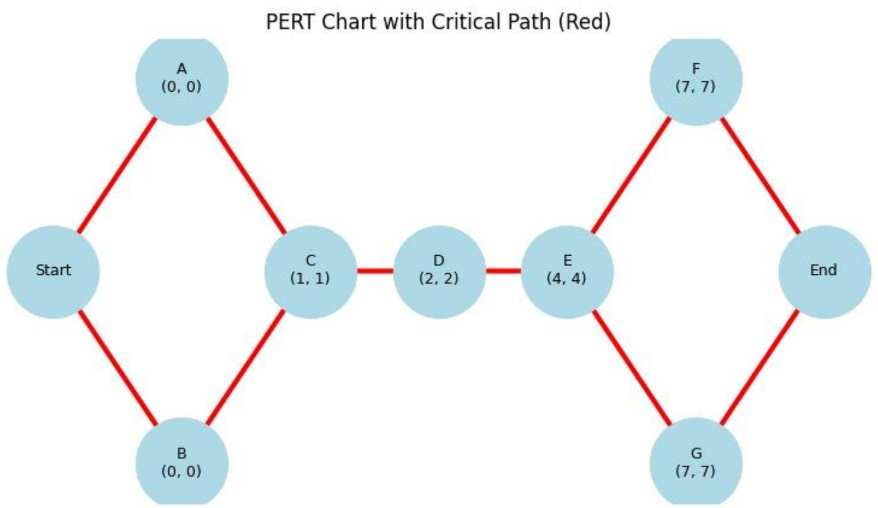
Schedule (6 Weeks)

Week	Activity
1	Requirements gathering and planning
2	System analysis and design
3	Implementation (core modules)
4	Implementation (complete prototype)
5	Testing and bug fixing
6	Documentation and final presentation

Milestones

- Planning (Week 1)
- Requirements (Week 1)
- Design (Week 2 - Week 3)
- Implementation (Week 3 – Week 5)
- Testing completed (Week 6)
- Final submission (Week 6)





7. Requirements Analysis and Design

Functional Requirements

The system must:

- Digitally register each student that attends an exam.
- Verify that the person writing the examination is the actual registered student.
- Track that the student submitted their examination script.
- Automatically count the students present for the exam in the venue.

Design Methods

- Use-case diagrams to show who interacts with the system and what actions do they perform.
- Flowcharts to show what happens first, next, and after that.
- Database design
- Interface mockups which will show what the user sees on the screen.

Tools

- Draw.io (Diagrams)
- VS Code
- Django framework

8. Implementation

Technologies

- Python
- Django
- SQLite (prototype database)
- HTML
- JavaScript

Modules to Implement

- Student verification module
- Attendance recording module
- Attendance count dashboard
- Admin report module

9. Testing

Test Environment

We will test on Laptop computers, Localhost Development Environment and on a simulated exam attendance data.

Testing Types

- Unit testing
- Integration testing
- User acceptance testing

Test Procedures

Test:

- Student verification
- Attendance recording
- Automatic counting accuracy
- Report of sum of students

10. Resources

Hardware

- Laptops
- Internet connection

Software

- VS Code
- Python
- GitHub
- Draw.io

Human Resources

- 4 Team Members
- Sample users for testing

11. Quality Assurance

Quality will be ensured through:

- Regular code review
- Weekly progress checks
- Testing after each module
- Validation against project requirements
- Final system evaluation

12. Changes

Changes will be managed through: Team approval before major changes, Git version control, documentation of all changes, restricting changes after Week 4 unless critical.